



SANDEEP KUMAR SINGH

Senior AWS Data Engineer

My Contact

✉ sandeepaotian@gmail.com

☎ 9681365422

📍 Pune

Skill

- Python
- Python Library – Pandas, Matplotlib, Scikit Learn etc
- SQL Performance Optimisation
- SQL, PostgreSQL, MongoDB
- Data Visualization
- Pyspark, SparkSql
- Data Warehouse – Amazon Redshift
- Data Modeling
- Data Analysis
- Oracle Database Management System
- AWS – Sagemaker, Athena, Glue, S3 bucket, Step Function, SQS, SNS
- ETL Pipeline
- GIT, Unix, SOS Scheduler
- Data Engineering

Education Background

- West Bengal University of Technology
B.Tech (Mechanical Engineering) – 2009–2013
- Kendriya Vidyalaya Patna
Higher Secondary (XII) – 2008
- Kendriya Vidyalaya Patna
Matriculation – 2006

Certifications

- AWS Certified Data Engineer Associate
- Data Science with Python from Data Brio Academy
- Machine Learning by Andrew Ng from Coursera
- MongoDB database from MongoDB University

About Me

Experienced Python Developer and **AWS Certified Data Engineer** with 8 years of experience in developing robust solutions, implementing end-to-end **data analytics** projects, creating **AWS pipelines** for data cleaning and transformation, and optimizing data models for impactful **data visualization**."

Professional Experience

Synechron | Senior AWS Data Engineer | Sep 2024 – Present

- Implemented scalable ETL pipelines using **AWS Glue, Lambda, and Step Functions** to ingest and transform structured and semi-structured data from core banking systems, APIs, and third-party sources.
- Built and maintained data lakes on **Amazon S3** using Lake Formation with fine-grained access control for regulatory compliance and secure analytics.
- leveraged AWS Glue **crawlers** and **AWS Glue Catalog** to register metadata and enable queryability through Amazon Athena and Redshift Spectrum.
- Developed and orchestrated serverless workflows using AWS Step Functions and integrated them with data quality checks using AWS Lambda and **SNS/SQS**.

ITC Infotech | Data Engineer | Feb 2023 – Present

- Designed and implemented complex **data models** for new applications and systems, resulting in a **improved query performance**.
- Designed and implemented end-to-end **data pipelines** to **collect, clean, and transform** raw data into usable formats for analysis using **AWS Glue**.
- Developed automated scripts and workflows to efficiently handle **data ingestion, processing, and storage**, removing manual effort using Python, **Pyspark**, SparkSql.
- Conducted thorough **data exploration and analysis** to identify trends, patterns, and anomalies, facilitating **strategic decision-making**.
- Designed and implemented **data warehouses** and data marts, optimizing storage and retrieval of large datasets.
- Utilized cloud platforms **AWS** for data storage and processing, including **S3, Athena, Glue, Redshift, lambda**.

Tata Consultancy Services | Data Engineer | Sep 2017 – Feb 2023

- Using **Python, PySpark, AWS Glue** and **SQL** to ingest, clean, and transform data from various sources.
- Utilized Python libraries such as **pandas** and PySpark to perform complex **data transformations** and ensure **data quality**.
- Implemented **data validation** and cleansing scripts to improve the accuracy and consistency of datasets.
- **Optimized SQL queries** for better performance and faster data retrieval.
- Developed and optimized complex PL/SQL procedures, functions, triggers, and packages to support business applications.
- Created and maintained database objects (tables, indexes, views) based on application needs.

Tata Consultancy Services | Design Engineer | Mar 2014 – Aug 2017

- Used CAD software to create models and simulations for use by manufacturing team.
- Selected manufacturing methods, fabrication and product designs with care for internal cost controls and government standards.